SPACE OPERATIONS · TAKE-HOME TEST

SpaceOps take-home test: anomaly detection

Seven satellites, alpha to golf: 72 h of telemetry, from 2019-05-30 10:00 to 2019-06-02 10:00 UTC, and one year of beta angles.

AT A GLANCE

Question 1	The data is loaded and cleaned (asynchronous rows, fill values, one glitch, outages), and the plots are in `plots/`.
Question 2	Wheel saturation, low voltage and gyro rates above 2 deg/s per satellite. On echo and golf, counting points understates the time spent in low voltage.
Question 3a	Two flocks: alpha, charlie and foxtrot; bravo, echo, golf and delta, whose orbit slowly drifts away from the other three.
Question 3b	delta loses attitude control and tumbles. echo no longer recharges its battery, a trend that bravo and golf follow later.
Question 4	Six detection functions, each on one telemetry channel. On the seven satellites they fire only on delta, echo and bravo.

1 Retrieve, filter and plot the data

Data format issues and how they are handled

1. **Asynchronous rows.** Each subsystem reports at its own rate (gyros ~10 Hz, wheels ~0.1 Hz, power every 1 to 90 s), and a row only carries what was sampled at that instant, so most cells are empty. Each channel is treated as its own time series, with its empty cells dropped.
2. **Fill values.** delta reports `bus_voltage = 0 V` and `low_voltage = 0` on every row, at the gyro rate. A powered satellite cannot read 0 V, and 0 V would raise the flag, so delta's power channels are treated as not telemetered.
3. **Glitch.** One foxtrot sample reads 2.08 V between two 8.30 V samples, with the flag at 0. The value is suspicious because it deviates from neighboring measurements, but that alone isn't enough to identify the error; we would need the hardware documentation. I've decided to remove it temporarily.
4. **Outages.** echo sends no packet at all for 18 h (2019-06-01 14:51 to 06-02 08:51 UTC), and bravo has two silences of about 30 min. Single channels have more holes. Plotted lines break over gaps longer than 10 min instead of bridging them.
5. **Beta file layout.** Unnamed index column, m/d/yyyy dates, and satellites in a different column order than in the rest of the data.
6. **Beta file dates.** The beta curves only match an orbit model once their dates are moved back by 32 days, the same for all satellites (residual 0.005 deg instead of about 2 deg). Comparisons between satellites are unaffected; the absolute beta values correspond to the Sun about a month earlier than the date shown.

Plots

All plots are in the `plots/` folder. I represented the ±2 deg/s threshold on the gyros (as mentioned in question 2.c) and the 8000 rpm saturation on the wheels.

Question	Files	Content
1a · one plot per satellite	`q1a_<satellite>.png`, 7 files	Four panels share the time axis: gyro rates, wheel speeds, bus voltage with the low-voltage periods shaded, and the low-voltage flag. Axis ranges are the same for every satellite, so the seven figures can be compared directly.
1b · one plot per channel	`q1b_<channel>.png`, 9 files	Each figure stacks one panel per satellite, with shared time and value scales.
1c · beta angle	`q1c_beta_angle.png`, 1 file	All seven satellites on one axis over the year of the beta file, with the 3-day telemetry window shaded.

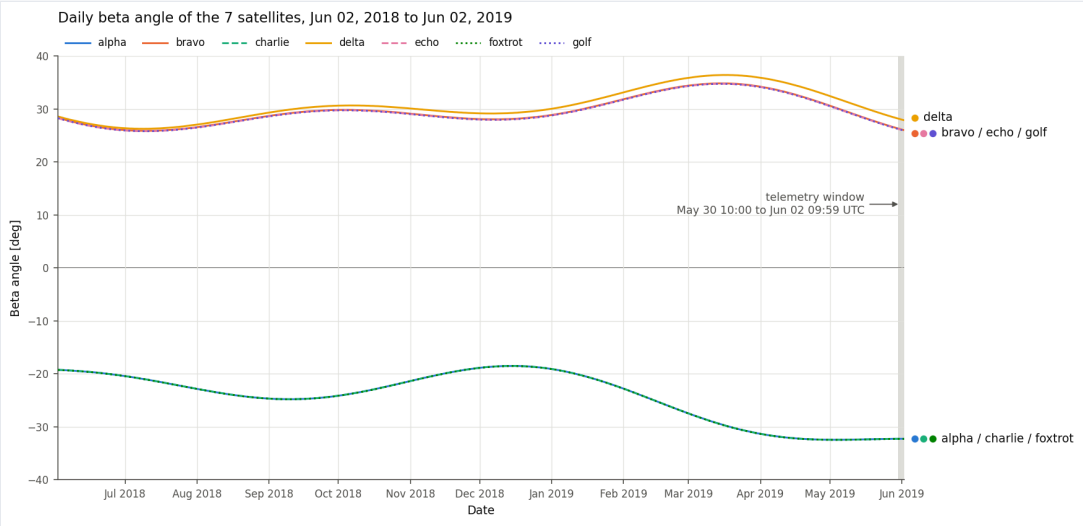


Figure 1. Beta angle of the seven satellites over the year of the beta file (`q1c_beta_angle.png`).

2 Share of the available telemetry points

Computed by `q2_percentages.py`.

Satellite	a. Wheel saturated	b. Low voltage (points)	b. Low voltage (time)	c. Gyro > 2 deg/s
alpha	1.81 %	0.00 %	0.00 %	0.43 %
bravo	2.80 %	**20.51 %**	**29.97 %**	0.71 %
charlie	1.55 %	0.00 %	0.00 %	0.17 %
delta	3.05 %	*n/a*	*n/a*	**5.88 %**
echo	3.13 %	**35.71 %**	**61.51 %**	1.01 %
foxtrot	1.28 %	0.00 %	0.00 %	0.30 %
golf	2.86 %	1.47 %	3.87 %	0.85 %

- Low voltage means the low_voltage flag is raised (`low_voltage = 1`). Sampling seems to depend on the satellite's state. For example, on echo and golf the flag is sampled every ~1 s while it is 0 but only every ~90 s while it is 1, so the share of points understates the time spent in low voltage.
- delta's power values seem corrupted, so they are considered not telemetered.

3 Flocks and anomalies

3a · Flocks: two flocks, delta drifts slightly within flock 2

Flock	Satellites	Beta angle over the year	Orbital period (altitude)
1	alpha, charlie, foxtrot	−19 to −32 deg, identical within 0.03 deg	93.6 to 93.7 min (~455 km)
2	bravo, echo, golf	+26 to +35 deg, identical within 0.09 deg	94.2 to 94.6 min (~490 km)
2	delta	close to bravo, echo and golf, but drifts away from +0.3 to +1.9 deg	~94.4 min

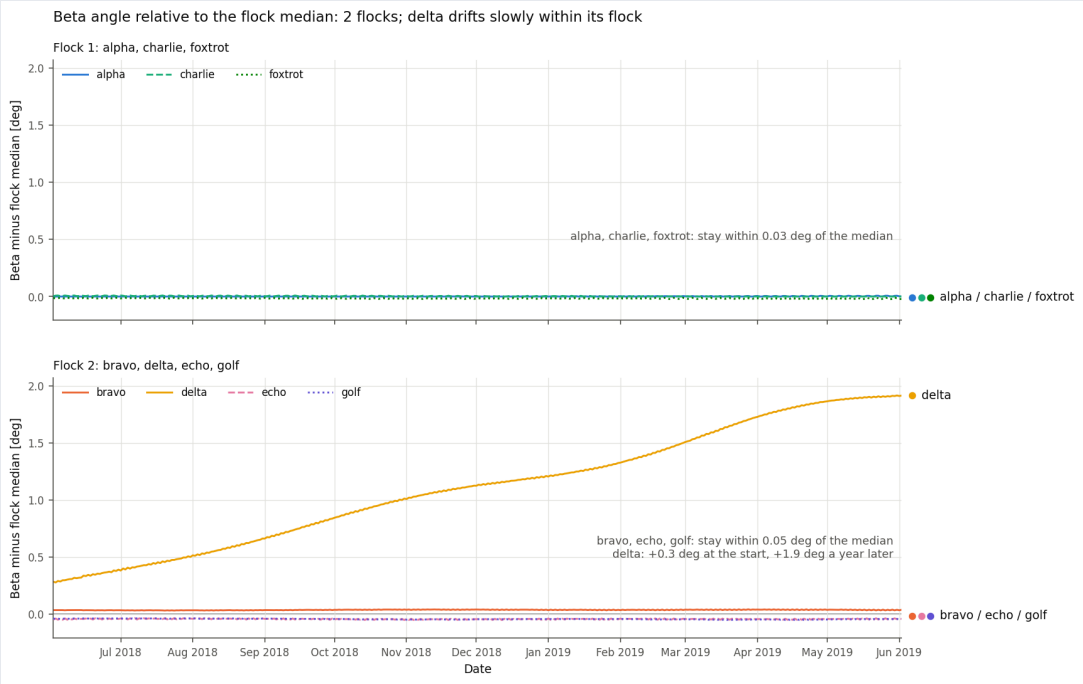


Figure 2. Beta angle relative to the flock median (`q3a_flocks.png`). Inside flock 1 the curves stay within 0.03 deg; in flock 2, delta slowly drifts away from bravo, echo and golf.

How the orbits are measured

Altitude, from the orbital period. The bus voltage repeats once per orbit (charge in sunlight, discharge in eclipse). Considering a circular Sun-Synchronous Orbit (SSO), based on the analysis of TLE data from satellites belonging to the PlanetScope constellation (source), the orbital period T can be used to estimate the orbital altitude:

$$T = 2\pi\sqrt{\frac{a^3}{\mu}} \quad \Rightarrow \quad h = \left(\mu\left(\frac{T}{2\pi}\right)^2\right)^{\frac{1}{3}} - R_{\text{Earth}}$$

where $R_{\text{Earth}} = 6378$ km, $\mu \approx 398\,600.44$ km^3/s^2, T is in seconds, a is the distance from the center of Earth and h the satellite altitude above the Earth's surface. delta has no voltage telemetry, and its wheel speeds give a period too noisy to compare it with bravo, echo and golf.

Orbital plane, from the beta angle. The beta angle changes as the Earth moves around the Sun and the orbital plane slowly turns around the Earth's axis (precession motion), caused by the Earth's slightly flattened shape. For a nearly circular orbit, the precession speed is proportional to $a^{-7/2} \cdot \cos\left(i\right)$, where $a = R_{\text{Earth}} + h$ and i is the orbital inclination. Thus, even small variations in altitude or inclination can lead to a drift in the beta angle, as the orbital plane precesses at slightly different rates (source).

Why does delta drift?

`q3a_delta_drift.png` presents several scenarios assuming different orbital altitudes and inclinations. Starting from the offset observed on the first day, an altitude about 10 km higher or an inclination about 0.04 deg lower than bravo's both follow the observed drift closely. With the size of the difference fitted (`q3a_delta_fit.png`), an inclination 0.036 deg lower matches the observation to 0.004 deg rms, better than an altitude 9.3 km higher (0.011 deg rms).

While the hypothesis is plausible, it cannot be confirmed based on the available data. One possible explanation is that delta was not injected into the same orbit as the other satellites. delta also exhibits unusual behavior in other telemetry datasets. For example, the reaction wheel speeds sometimes appear to remain fixed for extended periods, either intentionally as part of the spacecraft operations or because of a hardware anomaly. Such behavior could produce differential drag, but drag changes the altitude, not the inclination, so it is unlikely to explain this drift.

Additional telemetry would be required to investigate this hypothesis. Relevant data could include attitude-control modes, reaction-wheel activation commands, motor currents, telemetry validity flags, and attitude setpoints. These parameters would help determine whether the observed behavior results from an intentional operational strategy or from an underlying spacecraft anomaly. Orbital elements from around deployment (TLE history) would show whether the orbit difference dates from injection.

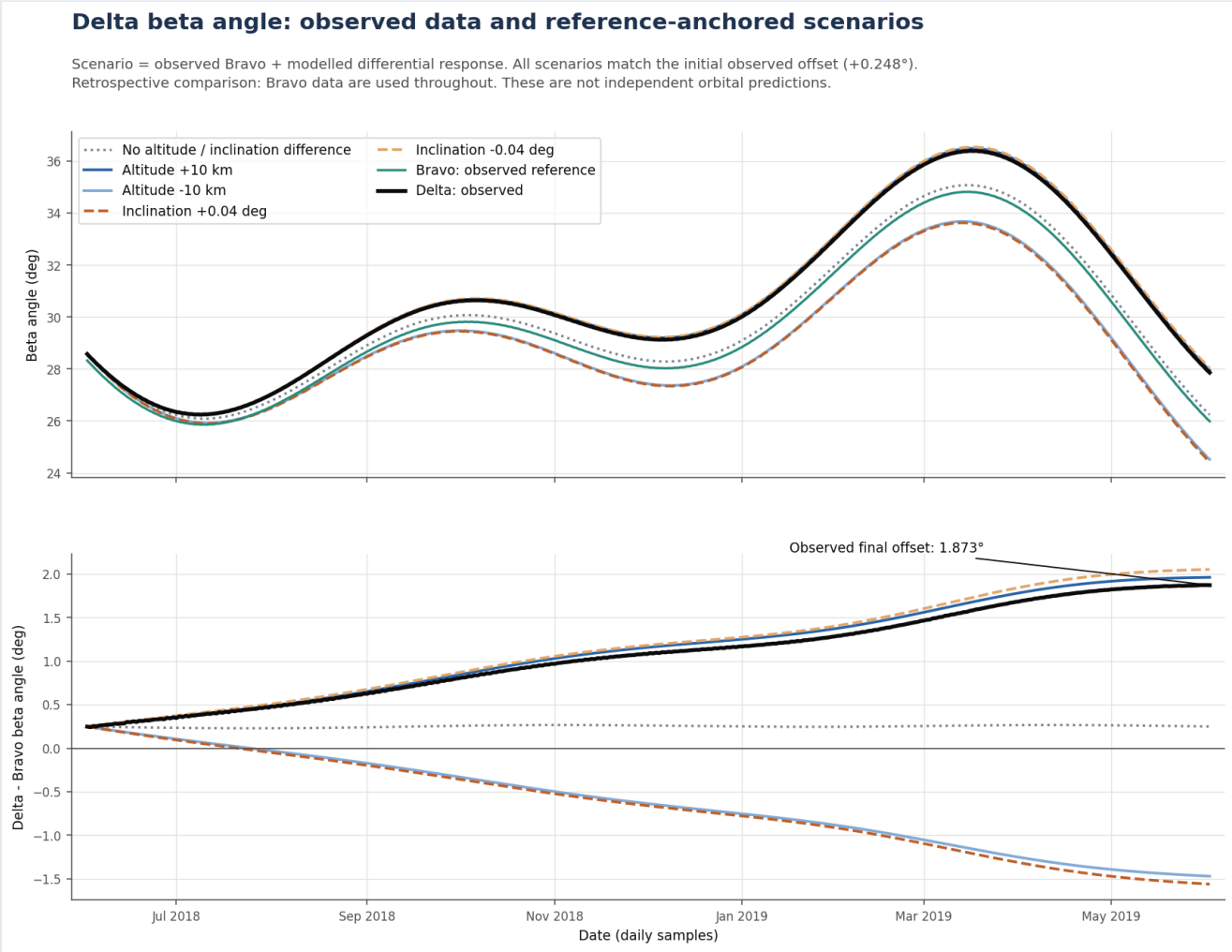


Figure 3. delta's beta angle against reference-anchored scenarios: observed bravo plus the modelled response to an altitude of ±10 km or an inclination of ±0.04 deg (`q3a_delta_drift.png`).

3b · Anomalies

1 · delta loses attitude control and tumbles

- Two of its four wheels (Y, Z) read 0 rpm for the whole 72 h, and wheel X half of the time. With fewer than three working wheels, the satellite cannot be controlled on three axes, so the remaining wheel S saturates for up to 35 min. On the other satellites, saturation never lasts more than about 2 min.
- On 2019-06-02 at 00:33 UTC all wheels drop to 0 rpm. The body rate then grows steadily, likely fed by the momentum the wheels stored as they spin down. It stays above 2 deg/s from 05:16 and reaches 4 deg/s by 09:00, still rising.
- A tumbling satellite cannot image, cannot point its antenna for downlinks, and cannot keep its solar panels on the Sun. The faster it spins, the harder the recovery. delta also has no power telemetry, so its battery cannot be watched.

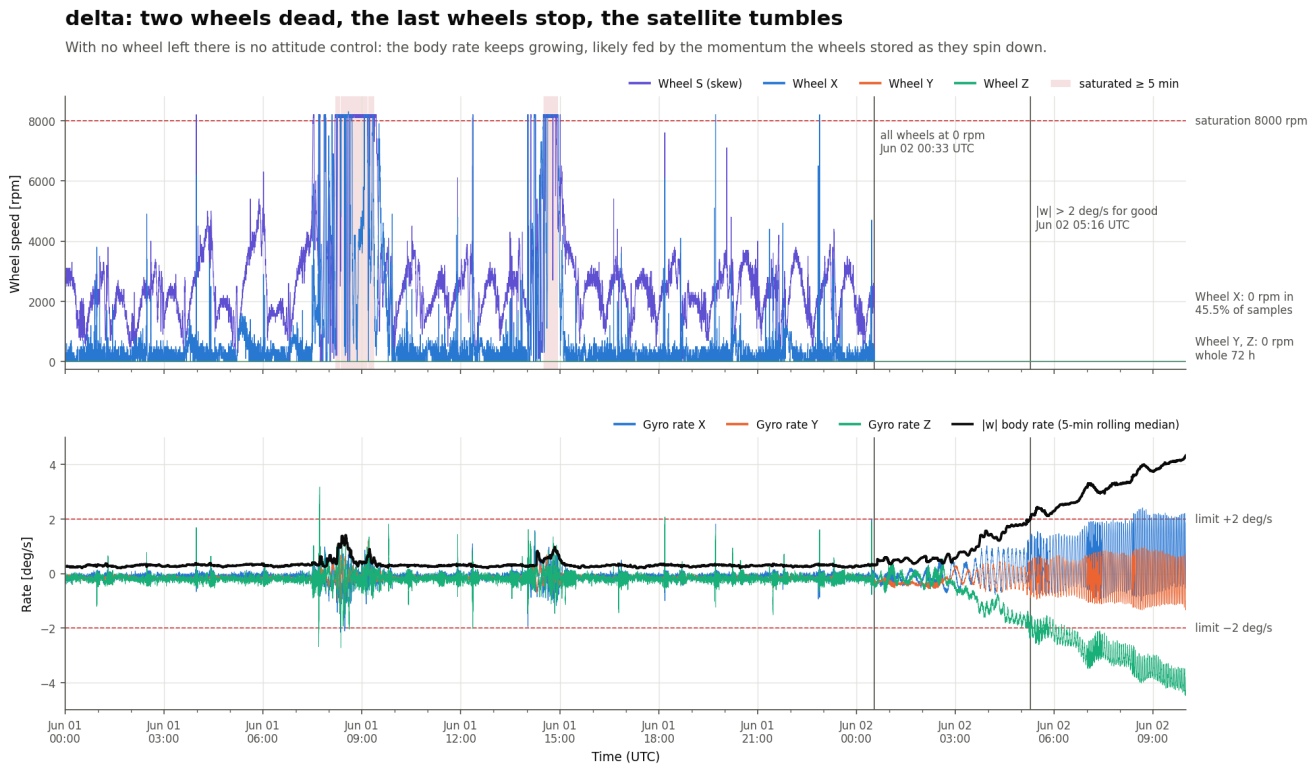


Figure 4. delta's wheel speeds and body rates over its last 34 h (`q3b_delta_tumbling.png`).

2 · echo no longer recharges its battery

- A healthy satellite reaches full charge (8.3 V) every orbit. For echo, the peak battery voltage drops to ~7.5 V within half a day.
- The low-voltage flag goes from 0% to 90% of the orbit within 3 orbits, indicating prolonged camera inhibition.
- Telemetry stops shortly afterwards: voltage data ends at 2019-06-01 12:54 UTC, all channels at 14:51 UTC, and telemetry resumes only 18 h later. This suggests a safe mode or reset triggered by undervoltage.
- The anomaly appears to affect flock 2 as a whole, with bravo and golf showing the same trend later. Flock 1 remains fully charged. Since both flocks experience similar eclipse durations, the cause is more likely related to battery ageing or operations than to the orbital environment. Other flock 2 satellites should be investigated.

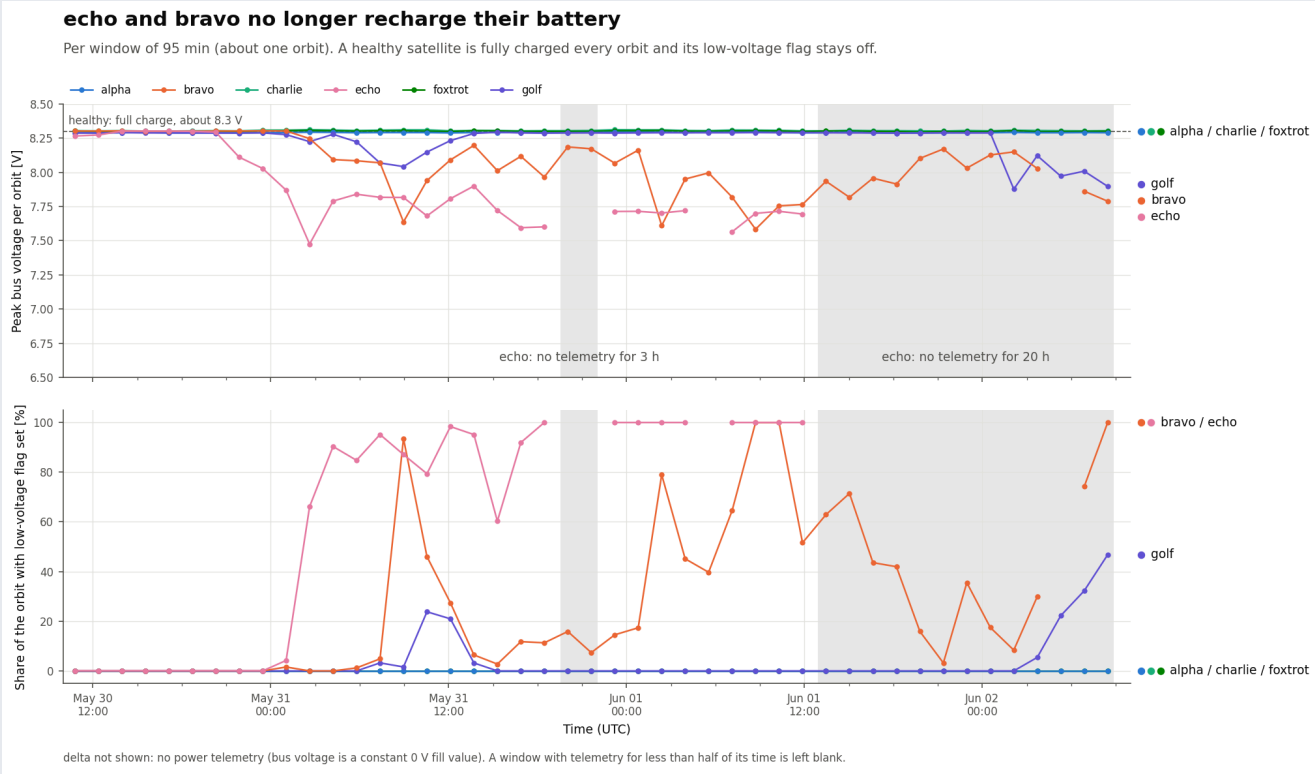


Figure 5. Peak bus voltage and share of the orbit with the low-voltage flag set, per orbit (`q3b_power.png`).

4 Detection functions

In `q4_detectors.py`, each function takes one telemetry channel, a pandas Series indexed by time (NaN epochs allowed), and returns True when the issue is present. A channel with no data returns False: missing telemetry is not evidence of the issue.

Function	Channel	Issue	True when	Worst case without the issue	With the issue
`is_wheel_stopped`	wheel_*	dead wheel (delta)	0 rpm without a break for ≥ 1 h	8 min	9 to 72 h
`is_wheel_saturated_too_long`	wheel_*	no momentum margin left (delta)	≥ 8000 rpm for ≥ 10 min	2.2 min	35 min
`is_tumbling`	gyro_*	attitude lost (delta)	5-min median of the absolute rate > 2 deg/s for ≥ 30 min	1.2 min	4 h
`is_battery_not_recharging`	bus_voltage	power-negative (echo, bravo)	peak per orbit < 8.1 V for ≥ 6 orbits in a row	3 orbits (golf)	11 to 24 orbits
`is_low_voltage_latched`	low_voltage	power-negative (echo, bravo)	flag at 1 without a break for more than one orbit (95 min)	42 min (golf)	4 to 7 h
`is_telemetry_lost`	any	silent satellite (echo)	no sample for ≥ 6 h	1 h (bravo)	18 h

Why there are no false positives

- The functions test sustained conditions, not single samples. The 2 deg/s and 8000 rpm limits of question 2 alone would fire on every satellite, since slews cross them for a minute or two.
- Each threshold has a physical meaning. A working wheel only touches 0 rpm while it reverses. A slew lasts minutes. A healthy battery is back to full charge after every sunlit phase, and one orbit always contains one.
- Applied to all seven satellites, the functions only fire on the anomalies of question 3b (printed by `main.py`):

Issue	alpha	bravo	charlie	delta	echo	foxtrot	golf
wheel stopped				**yes (4 wheels)**			
wheel saturated too long				**yes (S, X)**			
tumbling				**yes (Z)**			
battery not recharging		**yes**			**yes**		
low voltage latched		**yes**			**yes**		
telemetry lost					**yes**		

Limits

- golf's early power degradation remains below the detection thresholds (3 consecutive orbits below full charge and 42 min of continuous low-voltage flag in `q3b_power.png`), and is therefore not flagged. If this behaviour is considered operationally risky, the thresholds, particularly the 6-orbit criterion, could be reviewed. A lower-severity warning level could also be introduced.
- These functions are designed for post-analysis of recorded telemetry. In real operations, they would run continuously on a sliding window of recent data (e.g. the last 24 h). The `is_telemetry_lost()` detector would also need to be adapted, as it currently identifies telemetry outages by measuring the gap between consecutive samples. In a real-time stream, if telemetry stops completely, no subsequent sample is available for comparison, potentially delaying the alert until another mechanism checks the time elapsed since the last received sample.